Support Vector Regression: Risk Quadrangle Framework Case Studies

Project Report
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1 Abstract

The paper Support Vector Regression: Risk Quadrangle Framework [1] introduced the Support Vector Regression (SVR) is a widely used machine learning technique. The Risk Quadrangle (RQ) framework provides a unified interpretation of SVR through four interconnected stochastic functionals—error, regret, risk, and deviation—linked by a common statistic. Within this framework, the ε-SVR and ν-SVR formulations correspond to the minimization
the Vapnik error and the Conditional Value-at-Risk (CVaR) norm. These measures define quadrangles whose statistics equal the average of two symmetric conditional quantiles. Moreover, ν-SVR can be viewed as a form of Distributionally Robust Regression (DRR).

In this project, I replicated the theoretical results in Support Vector Regression: Risk Quadrangle Framework by Malandii and Uryasev [1]. I implemented the formulations in **Gurobi** [2] instead of the **PSG** solver used in the original case study. To validate the theoretical equivalence and computational performance, I conducted experiments on multiple datasets: a simple one-dimensional linear model with Laplace noise, a 10-dimensional uniform independent model, a 10-dimensional multivariate normal model with correlated features. The results confirm the consistency between the theoretical risk quadrangle interpretation and the optimization-based implementation. Finally, we discuss some potential methods that could enhance the results and also introduce how to build the non-linear SVR by applying the kernel-trick.

2 Introduction

Regression is a method used to approximate an unknown random variable Y by a function $\hat{f}$ of an observed random vector $\mathbf{X} = (X_1, ..., X_n)^T$. In simple terms, it tries to find a function $\hat{f}$ that best predicts Y. The accuracy of prediction is measured by minimizing the regression residual, denoted as $Z_f = Y - f(X)$. Norm serves as an error (e.g., L^1- regression, L^2 - regression) [3]. From a statistical perspective, the goal of regression is to estimate a conditional statistic $S(Y|X)$ of a random variable Y given X by finding a function $\hat{f}$, referred to as the best estimator. For example, L^1- regression estimates the conditional median, while L^2 - regression estimates the conditional mean.

In this paper, we focus on two general frameworks that are directly relevant to Support Vector Regression (SVR). First is the Vapnik–Chervonenkis (VC) theory which is a popular framework in machine learning [4]. It focuses on minimizing the expected loss $E[\ell(Z_f)]$ defined by a loss function ℓ. However, several important questions remain unresolved under the VC theory. In particular, it does not clearly specify which statistical quantity SVR is designed to estimate. The second framework defines the error measure $\mathcal{E}(Z_f)$ and uses RQ theory [5]links to other uncertainty measures:

risk, deviation, regret, and statistics.

To better understand how SVR relates to risk-based optimization, we first review the CVaR Optimization Formula [6], Dual CVaR Optimization Formula [7], and the Regular risk quadrangle [5].

CVaR is a risk measure that represents the expected loss beyond a given quantile level α.

Theorem 2.1 (CVaR Optimization Formula) *[6] For a random variable X and $\alpha \in (0,1)$,*

$$\bar{q}_\alpha(X) = \min_C \left\{ C + \frac{1}{1-\alpha}\, \mathbb{E}[X - C]_+ \right\}. \tag{1}$$

This formulation expresses CVaR as the minimum expected excess loss beyond the threshold C (Value-at-Risk or VaR), which is the quantile below which most losses occur. And $\mathbb{E}[X - C]_+$ measures the expected amount by which losses exceed C (this is also called regret).

Theorem 2.2 (Dual CVaR Optimization Formula) *[7]*

$$\mathbb{E}[X - x]_+ = \max_{\alpha \in [0,1]} (1-\alpha)(\bar{q}_\alpha(X) - x)\,, \tag{2}$$

and the set of maximizers for (2) is $\alpha \in [\mathbb{P}(X < x), \mathbb{P}(X \leq x)]$.

Definition 2.1 (Regular risk quadrangle) *[5] A quartet $(\mathcal{R}, \mathcal{D}, \mathcal{V}, \mathcal{E})$ of regular measures of risk, deviation, regret, and error is called a regular risk quadrangle if it satisfies the following relationships:*

$(Q1)$ *Error projection:* $\quad \mathcal{D}(X) = \min_C \{\, \mathcal{E}(X - C)\,\}$

$(Q2)$ *Regret formula:* $\quad \mathcal{R}(X) = \min_C \{\, C + \mathcal{V}(X - C)\,\}$

$(Q3)$ *Centerness:* $\quad \mathcal{R}(X) = \mathcal{D}(X) + \mathbb{E}[X], \qquad \mathcal{V}(X) = \mathcal{E}(X) + \mathbb{E}[X].$

Moreover, the quartet $(\mathcal{R}, \mathcal{D}, \mathcal{V}, \mathcal{E})$ is bound by the statistic $\mathcal{S}(X)$ that satisfies:

$(Q4)$ *Statistic:* $\quad \mathcal{S}(X) = \arg\min_C \{\, \mathcal{E}(X - C)\,\} = \arg\min_C \{\, C + \mathcal{V}(X - C)\,\}.$

Both ε-SVR and ν-SVR can be interpreted as minimizing specific risk and deviation measures. In this framework, SVR estimates the average of two symmetric conditional quantiles, meaning it can approximate different statistics such as the mean, median, or expectile by adjusting the parameters ε or ν. In addition, the authors derive a new dual formulation of the ν-SVR and discuss its nonlinear extension using the well-known "kernel trick." This formulation can be efficiently solved with optimization tools such as Gurobi or PSG. In this way, RQ theory [5] provides a new understanding of SVR and clarifies its underlying statistical meaning.

3 SVR Formulations

This section introduces two SVR formulations: ε-SVR and ν-SVR. The equivalence between these two models was established by Chang and Lin [2001] [8], showing that for any ν-SVR with parameter $\nu \in (0,1]$, there exists an $\varepsilon \geq 0$ such that the corresponding ε-SVR yields the same optimal solution.

3.1 The ε-SVR

The ε -insensitive loss function is defined as $L(\xi) = max[0, |\xi| - \varepsilon] = [|\xi| - \varepsilon]_+$. [4] This loss function penalizes only those data points that lie outside the ε-insensitive tube, that is, when the absolute value of the residual exceeds ε.
In Support Vector Regression: Risk Quadrangle Framework [1], it considers a linear regression with a training sample

$$X^l = (x_i, y_i)_{i=1}^l \,, \tag{3}$$

where $x_i \in R^n$ is a feature vector and $y_i \in R$ is a target value. The goal is to find a hyperplane $w^T x + b,\ (w, b) \in R^{n+1}$ that optimally fits the given training data.
Therefore, we have the ε-SVR introduced by Vapnik [1995] [4], which is to minimize with respect to w and b by taking the expectation of the ε-insensitive loss function plus a regularization term (to flatten the model) as follows:

$$\min_{w,\,b}\ \mathbb{E}\big[\,L(z(w,b))\big] + \frac{1}{2C}\|w\|_2^2\,. \tag{4}$$

if we consider $\lambda = \frac{1}{C}$, this problem is same as

$$\min_{w,\,b}\ \mathbb{E}\big[\,|z(w,b)| - \varepsilon\,\big]_+ + \frac{\lambda}{2}\|w\|_2^2\,. \tag{5}$$

Expand, where $z_i(w, b) = (y_i - w^T x_i - b)$, we get:

$$\min_{w,b}\ \frac{1}{N}\sum_{i=1}^{N}[\,|y_i - w^\top x_i - b| - \varepsilon\,]_+ + \frac{\lambda}{2}\|w\|_2^2\,, \tag{6}$$

where $\lambda > 0$, $\varepsilon > 0$, $\|\cdot\|_2$ denotes the ℓ_2-norm, and $[a]_+ = \max\{0, a\}$ is the positive part of a number $a \in \mathbb{R}$.

To solve this problem in Gurobi, we need to convert it into a nonlinear optimization problem formulation. Below is the complete formulation of the ε-SVR model:
Objective:

$$\min_{w,b}\ \frac{1}{N}\sum_{i=1}^{N} u_i + \frac{\lambda}{2}\|w\|_2^2 \tag{7}$$

Subject to:

$$\begin{aligned}
&u_i \geq 0, && i = 1, \ldots, N\\
&u_i \geq (y_i - w^T x_i - b) - \varepsilon, && i = 1, \ldots, N\\
&u_i \geq -(y_i - w^T x_i - b) - \varepsilon, && i = 1, \ldots, N
\end{aligned}$$

If we want to consider the percentage of points we want to use in this regression model, we add a parameter ν. Therefore, we introduce a new formula, the ν-SVR.

3.2 The ν-SVR

The ν-SVR introduced by Schölkopf et al. [2000] [9], can be formulated as follows

$$\min_{w,\,b}\ \mathbb{E}\big[\,L(z(w,b))\big] + \nu\varepsilon + \frac{1}{2C}\|w\|_2^2\,. \tag{8}$$

if we consider $\lambda = \frac{1}{C}$, this problem is same as

$$\min_{w,\,b,\,\varepsilon}\ \mathbb{E}[\,|z(w,b)| - \varepsilon\,]_+ + \nu\varepsilon + \frac{\lambda}{2}\|w\|_2^2\,. \tag{9}$$

where parameter $\nu \in (0, 1]$ controls the number of support vectors.
The minimum with respect to ε of the first two terms in the previous formula equals

$$\min_{\varepsilon} \ \mathbb{E}\big[\,|z(w,b)| - \varepsilon\,\big]_+ + \nu\varepsilon \ = \ \nu\,\bar{q}_{1-\nu} \ \equiv \ \langle\!\langle z(w,b)\rangle\!\rangle_{1-\nu}\,,$$

where $\bar{q}_{1-\nu}(\cdot)$ is the conditional value-at-risk (CVaR), and $\langle\!\langle\cdot\rangle\!\rangle_{1-\nu}$ is the CVaR norm. Therefore, ν-SVR can be reformulated as follows with $\nu = 1$ - α.

$$\min_{w,b} \quad \langle\!\langle z(w,b)\rangle\!\rangle_{\alpha} + \frac{\lambda}{2}\,\|w\|_2^2\ . \tag{10}$$

$$\min_{w,\,b}\,(1-\alpha)\,\bar{q}_{\alpha}\big(|z(w,b)|\big) + \frac{\lambda}{2}\|w\|_2^2\,. \tag{11}$$

Based on Theorem 2.1, we can reformulate this as follows:

$$\min_{w,\,b,\,C}\,(1-\alpha)\left(C + \frac{1}{(1-\alpha)N}\sum_{i=1}^{N}[\,|z(w,b)| - C\,]_+\right) + \frac{\lambda}{2}\|w\|_2^2\,. \tag{12}$$

To solve this problem in Gurobi, we need to convert it into a nonlinear optimization problem formulation, where $z_i(w,b) = (y_i - w^T x_i - b)$ Below is the complete formulation of the ν-SVR model after simplification:
Objective:

$$\min_{w,\,b,\,C}\,(1-\alpha)\,C + \frac{1}{N}\sum_{i=1}^{N} u_i + \frac{\lambda}{2}\|w\|_2^2\,, \tag{13}$$

Subject to:

$$\begin{aligned}
&u_i \geq 0, && i = 1,\ldots,N\\
&u_i \geq (y_i - w^T x_i - b) - C, && i = 1,\ldots,N\\
&u_i \geq -(y_i - w^T x_i - b) - C, && i = 1,\ldots,N
\end{aligned}$$

4 Risk Quadrangle Framework

In this section, we will introduce the CVaR Norm and the Risk Quadrangle framework, and examine how they relate to the ν-SVR problem.

Definition 4.1 (Scaled CVaR Norm) *[10] Let $X \in \mathcal{L}^1(\Omega)$ be a real-valued random variable. Then* scaled CVaR norm *of X with parameter $\alpha \in [0,1]$ is defined by*

$$\langle\!\langle X\rangle\!\rangle_{\alpha}^{S} = \bar{q}_{\alpha}\left(|X|\right)\ . \tag{14}$$

Definition 4.2 (Non-scaled CVaR Norm) *[10] Let $X \in \mathcal{L}^1(\Omega)$ be a real-valued random variable. Then* non-scaled CVaR norm *of X with parameter $\alpha \in [0,1)$ is defined by*

$$\langle\!\langle X\rangle\!\rangle_{\alpha} = (1-\alpha)\bar{q}_{\alpha}\left(|X|\right).\tag{15}$$

4.1 CVaR Norm Quadrangle

From the paper Support Vector Regression: Risk Quadrangle Framework [1], we know the following propositions:

Proposition 1 *For $\alpha \in [0,1)$, the error measure $\mathcal{E}_\alpha(X) = \langle\langle X \rangle\rangle_\alpha$ generates the following CVaR Norm Risk Quadrangle [11]:*

$$\mathcal{S}_\alpha(X) = \tfrac{1}{2}\big(q_{(1-\alpha)/2}(X) + q_{(1+\alpha)/2}(X)\big),$$

$$\mathcal{R}_\alpha(X) = \tfrac{1}{2}\big((1+\alpha)\,\bar{q}_{(1-\alpha)/2}(X) + (1-\alpha)\,\bar{q}_{(1+\alpha)/2}(X)\big),$$

$$\mathcal{D}_\alpha(X) = \tfrac{1}{2}\big((1+\alpha)\,\bar{q}_{(1-\alpha)/2}(X - \mathbb{E}[X]) + (1-\alpha)\,\bar{q}_{(1+\alpha)/2}(X - \mathbb{E}[X])\big) = \mathcal{R}_\alpha(X) - \mathbb{E}[X],$$

$$\mathcal{V}_\alpha(X) = \langle\langle X \rangle\rangle_\alpha + \mathbb{E}[X],$$

$$\mathcal{E}_\alpha(X) = \langle\langle X \rangle\rangle_\alpha .$$

Proposition 2 (Quantile Symmetric Average Quadrangle) *[1] Let $X \in L^2$, $0 \le \varepsilon < \frac{1}{2}(\operatorname{ess\,sup} X - \operatorname{ess\,inf} X)$, and $(\mathcal{R}_\alpha, \mathcal{D}_\alpha, \mathcal{V}_\alpha, \mathcal{E}_\alpha, \mathcal{S}_\alpha)$ be the CVaR Norm Quadrangle. Then the set*

$$\mathcal{A}_\varepsilon(X) = \big\{\, \alpha | \varepsilon \in \tfrac{1}{2}\big(q_{(1+\alpha)/2}(X) - q_{(1-\alpha)/2}(X)\big)\big\} \tag{16}$$

is nonempty, and the Vapnik error generates the following quadrangle:

$$\mathcal{S}_\varepsilon(X) = \bigcup_{\alpha \in \mathcal{A}_\varepsilon(X)} \mathcal{S}_\alpha(X),$$

$$\begin{aligned}
\mathcal{R}_\varepsilon(X) &= \mathcal{R}_\alpha(X) - (1-\alpha)\varepsilon, && \forall \alpha \in \mathcal{A}_\varepsilon(X),\\
\mathcal{D}_\varepsilon(X) &= \mathcal{D}_\alpha(X) - (1-\alpha)\varepsilon, && \forall \alpha \in \mathcal{A}_\varepsilon(X),\\
\mathcal{V}_\varepsilon(X) &= \mathbb{E}[|X| - \varepsilon]_+ + \mathbb{E}[X],\\
\mathcal{E}_\varepsilon(X) &= \mathbb{E}[|X| - \varepsilon]_+ = \textit{Vapnik error.}
\end{aligned}$$

The paper [1]proves that these two propositions are equivalent. Therefore, we can use Proposition 1 to reformulate the ν-SVR problem, allowing it to be interpreted within the Risk Quadrangle (RQ) framework.

4.2 The ν-SVR as Deviation Minimization Problems

Given a random vector of predictors $\mathbf{X} = (X_1, \ldots, X_n)^\top$ and a response variable Y, find a function f from a given class $\mathcal{F}$ that best approximates Y by solving the following optimization problem [1]:

$$\min_{f \in \mathcal{F}} \mathcal{E}(Z_f), \quad \text{where } Z_f = Y - f(\mathbf{X}) . \tag{17}$$

Solving the regression problem can be viewed as minimizing the deviation of the residuals, which represents the variability of prediction errors.

Define the regression residual as $Z_f = Y - w^\top \mathbf{X} - b$, without the intercept b, will denote as $\bar{Z}_f = Y - w^\top \mathbf{X}$, such that $Z_f = \bar{Z}_f - b$.
We can rewrite the equation as:

$$\min_{f \in \mathcal{F}} \min_{b} \mathcal{E}(\bar{Z}_f - b) \tag{18}$$

which is equal to:

$$\min_{f \in \mathcal{F}} \mathcal{D}(\bar{Z}_f) \tag{19}$$

Then, (w^*, b^*) is an optimal solution to (1) if and only if w^* is an optimal solution to

$$\min_{w} \mathcal{D}(\bar{Z}_f(w)) ,$$

and the optimal intercept b^* can be obtained from the statistic functional as

$$b^* \in \mathcal{S}(\bar{Z}_f(w^*)) = \arg\min_{b} \mathcal{E}\big(\bar{Z}_f(w^*) - b\big) .$$

Consequently, the ν-SVR model can be reformulated within the Risk Quadrangle (RQ) framework, where the deviation corresponds to a risk measure based on the Conditional Value-at-Risk (CVaR).

Let $Y - w^\top \mathbf{X} = \bar{Z}_f(w) \in \mathcal{L}^2(\Omega)$, $\lambda > 0$ be a regularization constant, and $(\mathcal{S}_\varepsilon, \mathcal{D}_\varepsilon)$ be the corresponding pair of the statistic and deviation from the CVaR Norm and Quantile Symmetric Average Quadrangles, respectively. The ν-SVR formula (10) can be reformulated as:

$$\min_w \ \mathcal{D}_\alpha(\bar{Z}_f(w)) + \frac{\lambda}{2}\|w\|_2^2\,, \tag{20}$$

which is equivalent to

$$\min_w \ \mathcal{R}_\alpha(\bar{Z}_f(w)) - \mathbb{E}[\bar{Z}_f(\omega)] + \frac{\lambda}{2}\|w\|_2^2\,, \tag{21}$$

where $\alpha \in [0, 1)$, and the optimal intercept satisfies $b^* \in \mathcal{S}_\alpha(\bar{Z}_f(w^*))$. After plugging in the formula in Risk Quadrangle, and with $z(w) : z_i(w) = y_i - w^\top x_i$.we can get:

$$\min_w \ \tfrac{1}{2}\big((1+\alpha)\,\bar{q}_{(1-\alpha)/2}(z(w)) + (1-\alpha)\,\bar{q}_{(1+\alpha)/2}(z(w))\big) - \frac{1}{N}\sum_{i=1}^{N} z(w) + \frac{\lambda}{2}\|w\|_2^2\,. \tag{22}$$

To solve this problem in Gurobi, we need to convert it into a nonlinear optimization problem formulation. Below is the complete formulation of the ν-SVR model as Deviation Minimization Problem after simplification:

Objective:

$$\min_{w,\,u,\,v,\,c_1,\,c_2} \ \tfrac{1}{2}\Big[(1+\alpha)\,D_1(w, c_1, u) + (1-\alpha)\,D_2(w, c_2, v)\Big] + \frac{\lambda}{2}\|\mathbf{w}\|_2^2 \tag{23}$$

Subject to:

$$\begin{aligned}
&D_1(w, c_1, u) = c_1 + \frac{2}{(1+\alpha)N}\sum_{i=1}^{N} u_i - \frac{1}{N}\sum_{i=1}^{N}(y_i - w^\top x_i) \\
&D_2(w, c_2, v) = c_2 + \frac{2}{(1-\alpha)N}\sum_{i=1}^{N} v_i - \frac{1}{N}\sum_{i=1}^{N}(y_i - w^\top x_i) \\
&u_i \geq 0, && i = 1, \ldots, N \\
&v_i \geq 0, && i = 1, \ldots, N \\
&u_i \geq (y_i - w^\top x_i) - c_1, && i = 1, \ldots, N \\
&v_i \geq (y_i - w^\top x_i) - c_2, && i = 1, \ldots, N
\end{aligned}$$

where $b^* = \frac{c1+c2}{2}$

In the preceding discussion, we have shown that the ε-SVR, ν-SVR, and the standard SVR formulation can be interpreted as equivalent representations of the Support Vector Regression problem within the Risk Quadrangle (RQ) framework. To further validate these theoretical connections, we design a series of case studies and numerical experiments to empirically compare their performance and verify the consistency of the results under different model specifications.

5 Case Study

In this project, the three optimization formulations discussed above were implemented in Python using the Gurobi [2] solver. We let N = 1000, $\lambda = \frac{1}{2N}$. The objective was to numerically validate the theoretical equivalence between the ε-SVR, ν-SVR, and their corresponding CVaR-based Risk

Quadrangle (RQ) formulations, as well as to evaluate their computational stability and generalization performance.

The performance of each formulation was examined using both artificially generated datasets. All simulations were conducted with a fixed random seed to ensure reproducibility. The artificial datasets were designed to represent diverse data-generating processes, and the specific configurations for each setup are described below.

After constructing the linear regression models, I conducted five-fold cross-validation using MSE to assess out-of-sample performance. The resulting prediction accuracy and computational time were then compared across all models.

At the end, we demonstrate the method result for the best alpha value. Following the Support Vector Regression: Risk Quadrangle Framework Case Studies [1], the optimal value of α in ν-SVR is selected through a cross-validation procedure. A grid of candidate α values is tested, and for each value, the model is trained and evaluated on validation folds. The Mean Squared Error (MSE) is used as the primary performance metric. The α^* that minimizes the cross-validated MSE is then adopted for final training, ensuring the best trade-off between model fit and generalization under the CVaR-based Risk Quadrangle formulation.

5.1 Generated Datasets

Dataset 1: One-Dimensional Linear Model with Laplace Noise.

$$N = 1000, \quad x_i \sim linspace(0, 1, N), \quad e_i \sim \text{Laplace}(0, 1), \quad y_i = x_i + e_i.$$

Dataset 2: Ten-Dimensional Independent Uniform Model.

$$N = 1000, \quad d = 10, \quad X_{ij} \sim \text{Uniform}(-1, 1), \quad e_i \sim \text{Laplace}(0, 1), \quad y_i = \sum_{j=1}^{10} X_{ij} + e_i.$$

Dataset 3: Ten-Dimensional Correlated Multivariate Normal Model.

$$\Sigma_{ij} = \rho^{|i-j|}, \quad \rho = 0.3, \quad d = 10, \quad X_i \sim \mathcal{N}(\mathbf{1}_d, \Sigma), \quad e_i \sim \text{Laplace}(0, 1), \quad y_i = \mathbf{1}_d^\top X_i + e_i.$$

5.2 Results

First, we analyze each dataset individually and compare the results across the three methods. After presenting these direct comparisons, we then report the outcomes of a 5-fold cross-validation procedure. After that, we use the MSE as a metric to turn the best alpha value, and then report the outcomes of a 5-fold cross-validation procedure.

5.2.1 Result: Analyze each dataset individually

In the 1-D linear model, we can see the result in Table 1.; all three methods produce essentially identical estimates for both the weight w^* and the intercept b^*. Their MSE values are also the same. This indicates that, in a simple one-dimensional setting with nearly symmetric residuals, the statistic $\mathcal{S}_\alpha(X)$ from the Risk Quadrangle deviation formulation coincides naturally with the intercept that minimizes the primal SVR loss. As a result, the deviation-based formulation perfectly reproduces the classical SVR solution in this case.

Dataset 1 (1-D Linear Model)			
Method	**w***	**b***	**MSE**
v-SVR	0.9322214591	0.02030369567	2.045072956
ε-SVR	0.9322214791	0.0203036873	2.045072956
v-SVR (Deviation)	0.9322215	0.02024874134	2.210625397

α = 0.6

Table 1. 1-D Linear Model.

In the 10-dimensional independent uniform model, we can see the result in Table 2.; the three methods again yield nearly identical weight vectors w^*, confirming that the deviation formulation accurately captures the slope direction of the model even in higher dimensions. Because the chosen parameter value $\alpha = 0.6$ produces symmetric quantiles that lie close to the distribution's central region, the statistic-based intercept b^* also aligns closely with the primal SVR intercept. Since the residual distribution is relatively symmetric and light-tailed, the statistic center and the SVR loss-based center nearly coincide, resulting in extremely similar MSE values across all three methods.

Dataset 2 (10-D Independent Uniform Model)			
Method	**w***	**b***	**MSE**
v-SVR	[0.99237928 0.94805879 0.84969811 1.05910101 0.89015261 0.92818 0.9697669 1.00455563 1.11931959 1.02146133]	0.04244817863	1.849724666
ε-SVR	[0.99237928 0.9480588 0.84969811 1.05910101 0.89015261 0.92818 0.96976689 1.00455563 1.11931959 1.02146133]	0.0438974359	1.849796329
v-SVR (Deviation)	[0.99237928 0.94805879 0.84969811 1.05910101 0.89015261 0.92818 0.96976689 1.00455563 1.11931959 1.02146133]	0.04462147481	1.849833704

α = 0.6

Table 2. 10-D Independent Uniform Model.

In the correlated 10-dimensional multivariate normal model, we can see the result in Table 3.; all three methods again recover a similar weight vector w^*, showing that the deviation formulation remains stable for estimating the slope coefficients. The intercept estimates b^* also align closely across the three approaches, indicating that the statistic used in the deviation-based formulation provides a stable and consistent center for the residual distribution under this data-generating process. As a result, the predictive performance, reflected in nearly matching MSE values, is effectively equivalent across ν-SVR (primal), ε-SVR (primal), and ν-SVR (primal, Deviation).

Dataset 3 (10-D Correlated Multivariate Normal Model)			
Method	**w***	**b***	**MSE**
v-SVR	[0.90997602 1.02681664 1.10474237 0.91188512 0.93126511 1.07937217 0.98902262 0.88491053 0.95427641 1.01383111]	0.299071437	1.699229482
ε-SVR	[0.90997602 1.02681664 1.10474237 0.91188512 0.93126511 1.07937217 0.98902262 0.88491053 0.95427641 1.01383111]	0.300148652	1.699347316
v-SVR (Deviation)	[0.90997597 1.0268166 1.10474233 0.91188507 0.93126493 1.07937218 0.98902253 0.88491062 0.95427635 1.01383111]	0.3003751367	1.699372357

α = 0.6

Table 3. 10-D Correlated Multivariate Normal Model.

5.2.2 Result: Analyze each dataset using 5-fold cross-validation

Across the five folds, in the 1-D linear model, we can see the result in the Table 4.; all three methods show very similar estimates for both the weight w^* and the intercept b^*. Their MSE values are also very close. These findings indicate that, in a simple one-dimensional setting, the three approaches behave equivalently, yielding virtually indistinguishable parameter estimates and predictive performance.

Dataset 1 (1-D Linear Model)			
Method	**w***	**b***	**MSE**
v-SVR	0.93638	0.019908	2.051889
ε-SVR	0.932935	0.021959	2.051782
v-SVR (Deviation)	0.93638	0.019946	2.051888

α = 0.6
5-fold

Table 4. 1-D Linear Model (5-fold Cross Validation).

Across the five folds, in the 10-dimensional independent uniform model, we can see the result in Table 5.; only slight differences in both the weight w^* and the intercept b^*. Their MSE values are also very close. Their cross-validated MSE values are also nearly identical. These findings indicate that, in this 10-dimensional setting, the three approaches behave equivalently, yielding virtually indistinguishable parameter estimates and predictive performance.

Dataset 2 (10-D Independent Uniform Model)			
Method	**w***	**b***	**MSE**
v-SVR	[0.99471091, 0.96662609, 0.84596596, 1.07343763, 0.90847347, 0.94277979, 0.96903634, 0.99818689, 1.10182263, 1.03471299]	0.04492669451	1.881848809
ε-SVR	[0.994711, 0.966626, 0.845966, 1.073438, 0.908473, 0.94278 , 0.969036, 0.998187, 1.101823, 1.034713]	0.0469339462	1.881910658
v-SVR (Deviation)	[0.99470833, 0.96663286, 0.84596762, 1.07344041, 0.90847094, 0.94277857, 0.96903067, 0.99818192, 1.10182867, 1.03471227]	0.04763483702	1.881956566

α = 0.6
5-fold

Table 5. 10-D Independent Uniform Model (5-fold Cross Validation).

Across the five folds, in the 10-dimensional multivariate normal model, we can see the result in Table 7.; only slight differences in both the weight w^* and the intercept b^*. Their MSE values are also very close. Their cross-validated MSE values are also nearly identical. These findings indicate that, in this 10-dimensional setting, the three approaches behave equivalently, yielding virtually indistinguishable parameter estimates and predictive performance.

Dataset 3 (10-D Correlated Multivariate Normal Model)			
Method	**w***	**b***	**MSE**
v-SVR	[0.92334698, 1.02061429, 1.10503784, 0.90799611, 0.93360655, 1.08005717, 1.00334024, 0.8800612 , 0.94859654, 1.02050976]	0.2614501395	1.734798589
ε-SVR	[0.923347, 1.020614, 1.105038, 0.907996, 0.933607, 1.080057, 1.00334 , 0.880061, 0.948597, 1.02051]	0.2630774718	1.73488388
v-SVR (Deviation)	[0.92334696, 1.02061439, 1.10503798, 0.90799606, 0.93360654, 1.08005718, 1.00334021, 0.88006118, 0.94859655, 1.02050982]	0.2636470668	1.734926741

α = 0.6
5-fold

Table 6. 10-D Correlated Multivariate Normal Model (5-fold Cross Validation).

5.2.3 Result: Analyze each dataset using 5-fold cross-validation with the Best Alpha value

We found that alpha = 0.7 gave the smallest MSE. Across the five folds, in the 1-D linear model, we can see the result in Table 7.; all three methods show very similar estimates for both the weight w^* and the intercept b^*. Their MSE values are also very close. These findings indicate that, in a simple one-dimensional setting, the three approaches behave equivalently, yielding virtually indistinguishable parameter estimates and predictive performance. The MSE results show the model is slightly better than the model with alpha = 0.6.

Dataset 1 (1-D Linear Model)			
Method	**w***	**b***	**MSE**
v-SVR	1.029328	-0.021781	2.048593
ε-SVR	0.999363	0.0018	2.047778
v-SVR (Deviation)	1.029328	-0.021598	2.048623

best α = 0.7

5-fold

Table 7. 1-D Linear Model (5-fold Cross Validation with best alpha = 0.1).

We found that alpha = 0.1 gave the smallest MSE. Across the five folds, in the 10-dimensional independent uniform model, we can see the result in Table 8.; only slight differences in both the weight w^* and the intercept b^*. Their MSE values are also very close. Their cross-validated MSE values are also nearly identical. These findings indicate that, in this 10-dimensional setting, the three approaches behave equivalently, yielding virtually indistinguishable parameter estimates and predictive performance. The MSE results show the model is slightly better than the model with alpha = 0.6.

Dataset 2 (10-D Independent Uniform Model)			
Method	**w***	**b***	**MSE**
v-SVR	[0.99085911, 0.9408 , 0.92459405, 1.09321972, 0.98189299, 0.98484623, 1.00847683, 1.005318 , 1.02291553, 1.0289486]	0.04189154242	1.862021594
ε-SVR	[0.990859, 0.9408 , 0.924594, 1.09322 , 0.981893, 0.984846, 1.008477, 1.005318, 1.022916, 1.028949]	0.04450701332	1.862168687
v-SVR (Deviation)	[0.99085913, 0.94080001, 0.92459406, 1.09321972, 0.98189299, 0.98484622, 1.00847683, 1.00531799, 1.02291552, 1.02894859]	0.04513753748	1.862188264

best α = 0.1

5-fold

Table 8. 10-D Independent Uniform Model (5-fold Cross Validation with best alpha = 0.1).

We found that alpha = 0.5 gave the smallest MSE. Across the five folds, in the 10-dimensional multivariate normal model, we can see the result in Table 9.; only slight differences in both the weight w^* and the intercept b^*. Their MSE values are also very close. Their cross-validated MSE values are also nearly identical. These findings indicate that, in this 10-dimensional setting, the three approaches behave equivalently, yielding virtually indistinguishable parameter estimates and predictive performance. The MSE results show the model is slightly better than the model with alpha = 0.6.

Dataset 3 (10-D Correlated Multivariate Normal Model)			
Method	**w***	**b***	**MSE**
v-SVR	[0.92962045, 1.03178477, 1.11446451, 0.90791151, 0.96272689, 1.08410801, 0.98173674, 0.91069317, 0.97715422, 0.99077849]	0.1821072946	1.71693907
ε-SVR	[0.92962 , 1.031785, 1.114465, 0.907912, 0.962727, 1.084108, 0.981737, 0.910693, 0.977154, 0.990778]	0.1841461727	1.720538194
v-SVR (Deviation)	[0.92962044, 1.03178476, 1.11446452, 0.90791151, 0.96272689, 1.08410805, 0.98173676, 0.91069313, 0.97715425, 0.99077848]	0.1848901365	1.720550538

best α = 0.5

5-fold

Table 9. 10-D Correlated Multivariate Normal Model (5-fold Cross Validation with best alpha = 0.5).

5.2.4 Conclusion

This paper replicated the theoretical results from Support Vector Regression: Risk Quadrangle Framework [1], formulated ν-SVR, ε-SVR, and ν-SVR within the structure of the Risk Quadrangle (RQ). This representation highlights a deeper connection between machine learning and quantitative

risk management, showing that classical SVR models can be interpreted as optimization problems over risk measures and deviation statistics.

The theoretical analysis demonstrates that SVR is an asymptotically unbiased estimator of the average of two symmetric conditional quantiles. Building on this result, we show that the intercept $b*$ in the RQ formulation corresponds to the statistic component of the quadrangle, while the slope vector w^* is jointly determined through the CVaR-Norm deviation. Through multiple stochastic case studies and five-fold cross-validation, we empirically verified the near-equivalence between ν-SVR, ε-SVR, confirming that both methods recover almost identical parameter estimates and predictive performance across different data-generating processes—including 1-D linear, 10-D independent uniform, and 10-D correlated multivariate Normal models.

6 Improvement

6.1 Dual Formulation and Kernelization for Non-linear SVR

As discussed earlier, the paper [1] also demonstrates that the primal SVR problem can be reformulated as a dual optimization problem. This dual representation enables the use of the *kernel trick*, which extends SVR from a linear to a nonlinear setting by implicitly mapping the input data into a high-dimensional feature space.

Dual Formulation of SVR. Let $\nu = 1 - \alpha$. The primal problem can be written as:

$$\begin{aligned} \min_{w,\, b,\, z} \quad & C(1-\alpha)\, \langle\!\langle z \rangle\!\rangle_{\alpha}^{S} + \frac{1}{2}\|w\|_2^2, \\ \text{s.t.} \quad & z = y - \hat{\mathbf{X}} w - \mathbf{1}_l b, \end{aligned} \tag{24}$$

where $\mathbf{1}_l = (1, \ldots, 1)^\top \in \mathbb{R}^l$, $y = (y_1, \ldots, y_l)^\top$, and $\hat{\mathbf{X}} = (\mathbf{x}_1^\top, \ldots, \mathbf{x}_l^\top) \in \mathbb{R}^{l \times n}$.

By introducing a kernel function, the problem can be transformed into the following dual formulation:

$$\begin{aligned} \max_{\boldsymbol{\mu}} \quad & \boldsymbol{\mu}^\top \mathbf{y} - \frac{1}{2}\, \boldsymbol{\mu}^\top \hat{\mathbf{X}} \hat{\mathbf{X}}^\top \boldsymbol{\mu}, \\ \text{s.t.} \quad & \|\boldsymbol{\mu}\|_1 \le C(1-\alpha), \quad \|\boldsymbol{\mu}\|_\infty \le \frac{C}{l}, \\ & \boldsymbol{\mu}^\top \mathbf{1}_l = 0 \end{aligned} \tag{25}$$

with $K = \hat{\mathbf{X}} \hat{\mathbf{X}}^\top$:

$$\begin{aligned} \max_{\boldsymbol{\mu}} \quad & \boldsymbol{\mu}^\top \mathbf{y} - \frac{1}{2}\, \boldsymbol{\mu}^\top K \boldsymbol{\mu}, \\ \text{s.t.} \quad & \|\boldsymbol{\mu}\|_1 \le C(1-\alpha), \quad \|\boldsymbol{\mu}\|_\infty \le \frac{C}{l}, \\ & \boldsymbol{\mu}^\top \mathbf{1}_l = 0 \end{aligned} \tag{26}$$

To solve this problem in Gurobi, we need to convert it into a nonlinear optimization problem formulation. Below is the complete formulation of the Dual Formulation after simplification:

Objective:

$$\max_{\boldsymbol{\mu},\, \boldsymbol{\lambda}} \; \mu^\top \mathbf{y} - \frac{1}{2}\, \boldsymbol{\mu}^\top K \boldsymbol{\mu} \tag{27}$$

Subject to:

$$\begin{aligned}
&\mathbf{1}_N^\top \boldsymbol{\mu} = 0, \\
&\mathbf{1}_N^\top \boldsymbol{\lambda} \leq cN(1-\alpha), \\
&\boldsymbol{\lambda} \geq \boldsymbol{\mu}, \\
&\boldsymbol{\lambda} \geq -\boldsymbol{\mu}, \\
&-c \leq \boldsymbol{\mu}_i \leq c, \quad i = 1, \ldots, N.
\end{aligned}$$

7 Graphs

The following plots visualize the three models' fitted predictions y overlaid on the observed data. Despite the large amount of noise in the observations, the SVR solution recovers the underlying linear trend, which appears as a nearly horizontal fitted line across the cloud of points. This demonstrates that the model successfully identifies the linear relationship embedded in the data.
Finally, the last plot illustrates the SVR fit specifically for the one-dimensional linear model, providing a clearer visualization of how the ε-insensitive "tube" operates and how support vectors form around the boundary.

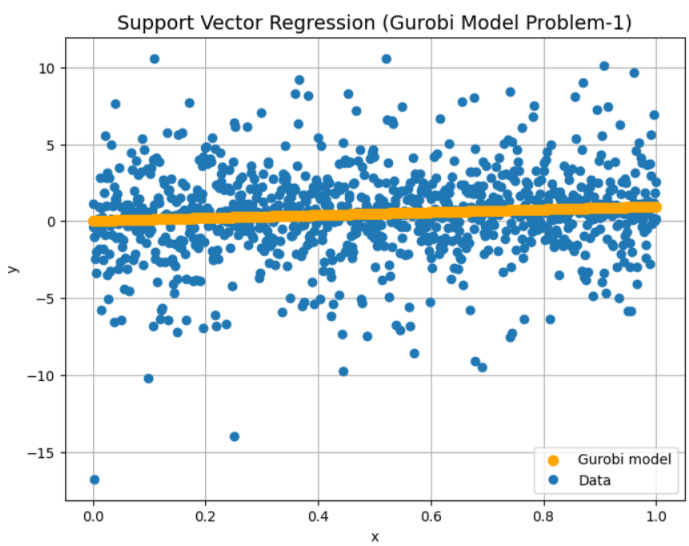

Figure 1. 1-D Linear Model

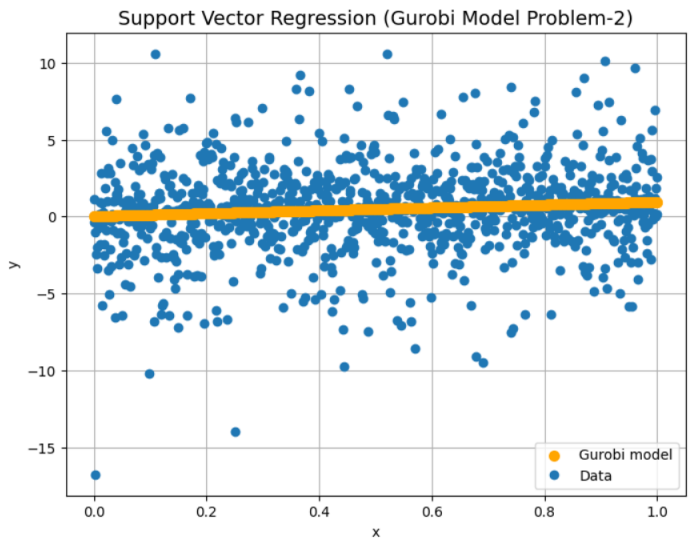

Figure 2. 10-D Independent Uniform Model

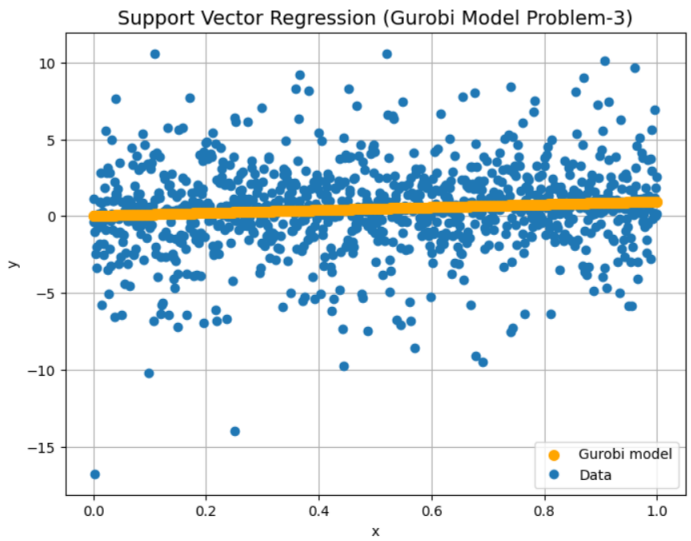

Figure 3. 10-D Correlated Multivariate Normal Model

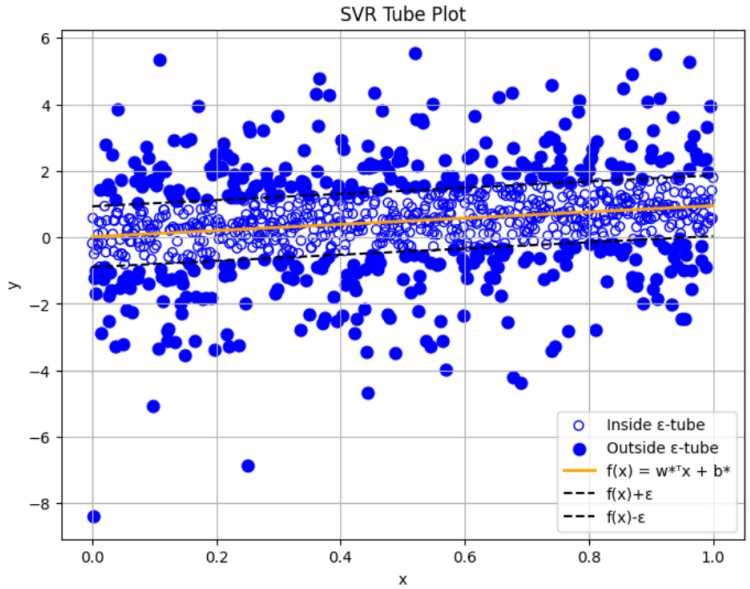

Figure 4. 1-D Linear Model (ε-SVR Plot)

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